ClipSAT : Contest Math & Olympiad AMC 10/12, Kangaroo & Mawhiba

Unit 4.01: Bijective Counting, Stars & Bars, PIE & Pigeonhole Name: _____

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 12x + 50 in the domain of Bijective Counting, Inclusion-Exclusion & Pigeonhole. By completing the square f_1x = x - h^2 + k , find the sum k + f_18 .

[ ] A: 38
[ ] B: 18
[ ] C: 30
[ ] D: 32

Q2. A mathematical model in Bijective Counting, Inclusion-Exclusion & Pigeonhole is defined by the cubic function y_2x = 2x^3 - 6x^2 + 15x - 11 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 33
[ ] B: 69
[ ] C: 18
[ ] D: 21

Q3. In Bijective Counting, Inclusion-Exclusion & Pigeonhole, two displacement vectors are given by mathbf{u}_{3} = langle 6, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[ ] A: -2
[ ] B: 98
[ ] C: 100
[ ] D: 210

Q4. The linear growth profile shown in the diagram for Bijective Counting, Inclusion-Exclusion & Pigeonhole follows an arithmetic progression with first term a_1 = 7 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 91
[ ] B: 825
[ ] C: 735
[ ] D: 1365

Q5. In Bijective Counting, Inclusion-Exclusion & Pigeonhole, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 24x,mathrm{d}x .

[ ] A: 120
[ ] B: 120 + 1
[ ] C: 49
[ ] D: 96

Q6. In Bijective Counting, Inclusion-Exclusion & Pigeonhole, consider the quadratic function f_6x = x^2 - 14x + 135 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_614 .

[ ] A: 221
[ ] B: 228
[ ] C: 135
[ ] D: 214

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Q7. The graph illustrates the cubic model y_7x = 2x^3 - 7x^2 + 33x - 18 studied in Bijective Counting, Inclusion-Exclusion & Pigeonhole. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[ ] A: 417
[ ] B: 305
[ ] C: 272
[ ] D: 67

Q8. In Bijective Counting, Inclusion-Exclusion & Pigeonhole, two displacement vectors are given by mathbf{u}_{8} = langle 7, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 21
[ ] B: 98
[ ] C: 119
[ ] D: 189